

① Motivation

Core Problem:

Real-world Mismatch: Conventional MARL/Meta-RL assumes synchronous, stationary task inputs. Temporal Misalignment: Actual human instructions (speech, visual cues) are asynchronous and bursty, conflicting with continuous robotic control processes.

Key Challenge: Aligning irregular, multimodal inputs with **continuous multi-agent decision-making** to achieve millisecond-level real-time responsiveness.

② Introduction

Background:

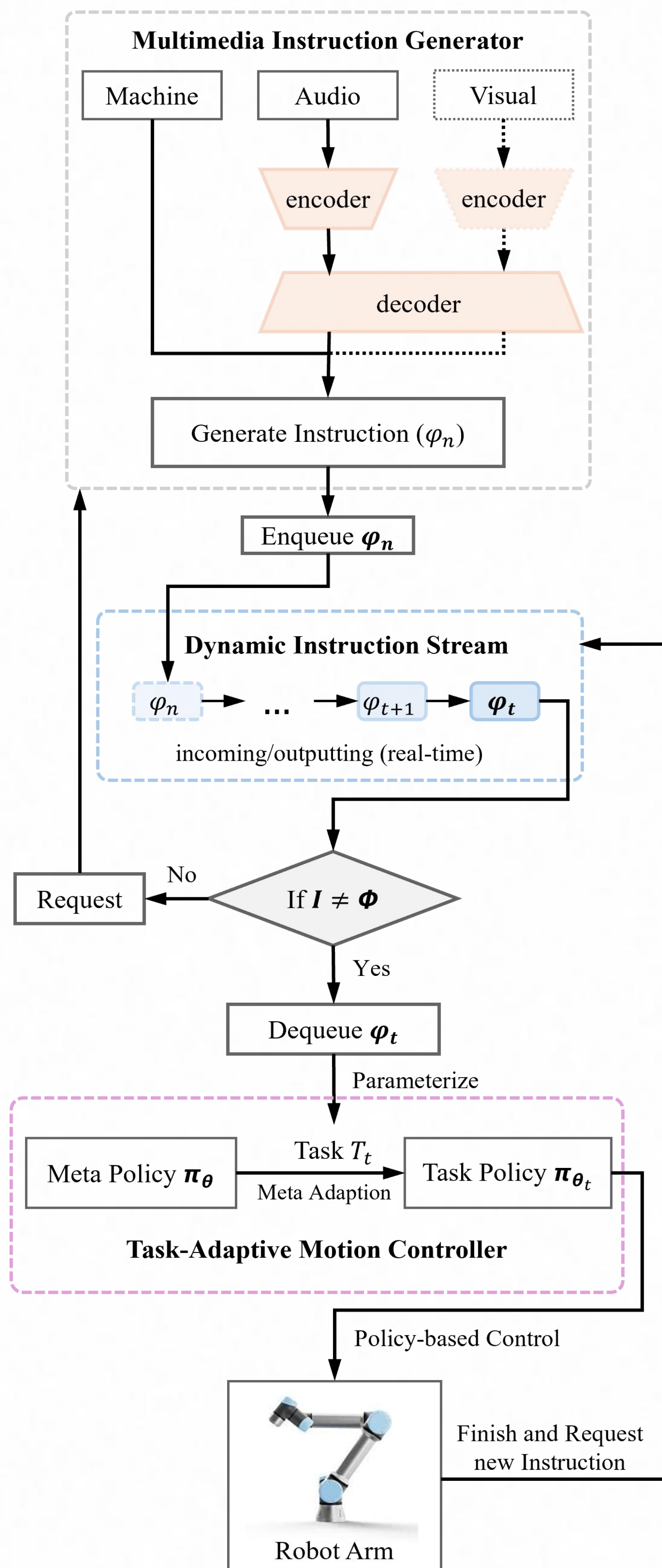
The Bottleneck: While models like SayCan and RT-2 advance embodied decision-making, they fail to handle the dynamic, asynchronous nature of real-world multimedia instruction streams.

Our Solution:

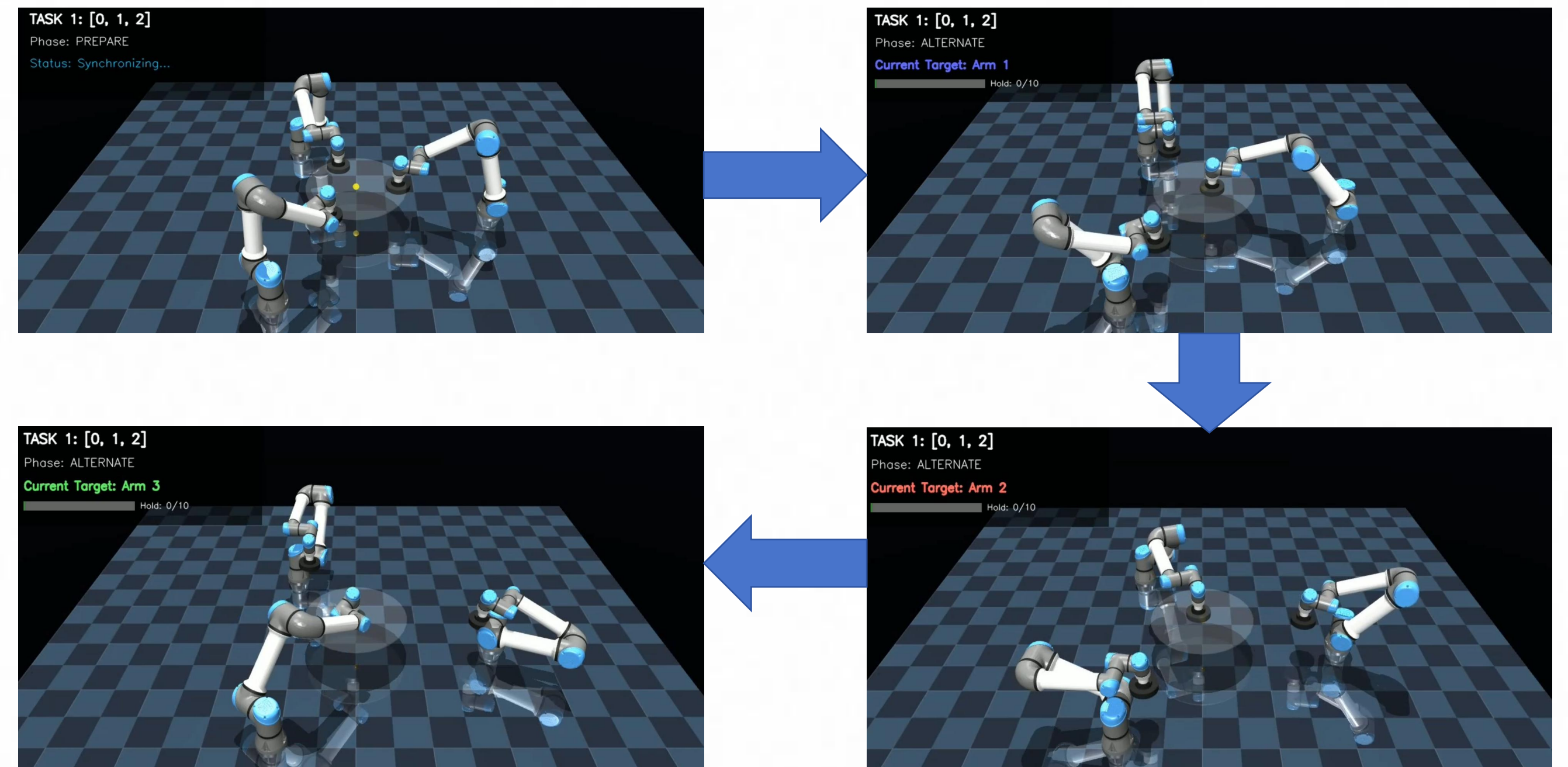
- 1.VA-FastNavi-MARL Unified Alignment:** Maps audio, visual, and machine inputs into a single modality-agnostic latent space.
- 2.Asynchronous Processing:** Uses a Dynamic Instruction Queue to bridge bursty human commands with continuous robot control. Fast Meta-
- 3.Adaptation:** Leverages MAML+SAC to instantly adapt to unseen instructions via minimal gradient steps.

③ VA-FastNavi-MARL Methods

Our framework maps asynchronous multimodal inputs into a unified latent space via cross-modal encoding. A dynamic FIFO queue aligns bursty human commands with continuous control loops. Using MAML and SAC, the meta-policy π_θ rapidly adapts to each instruction φ_t , generating a task-specific policy π_{θ_t} via minimal gradient updates for real-time multi-arm control.



④ Main Results

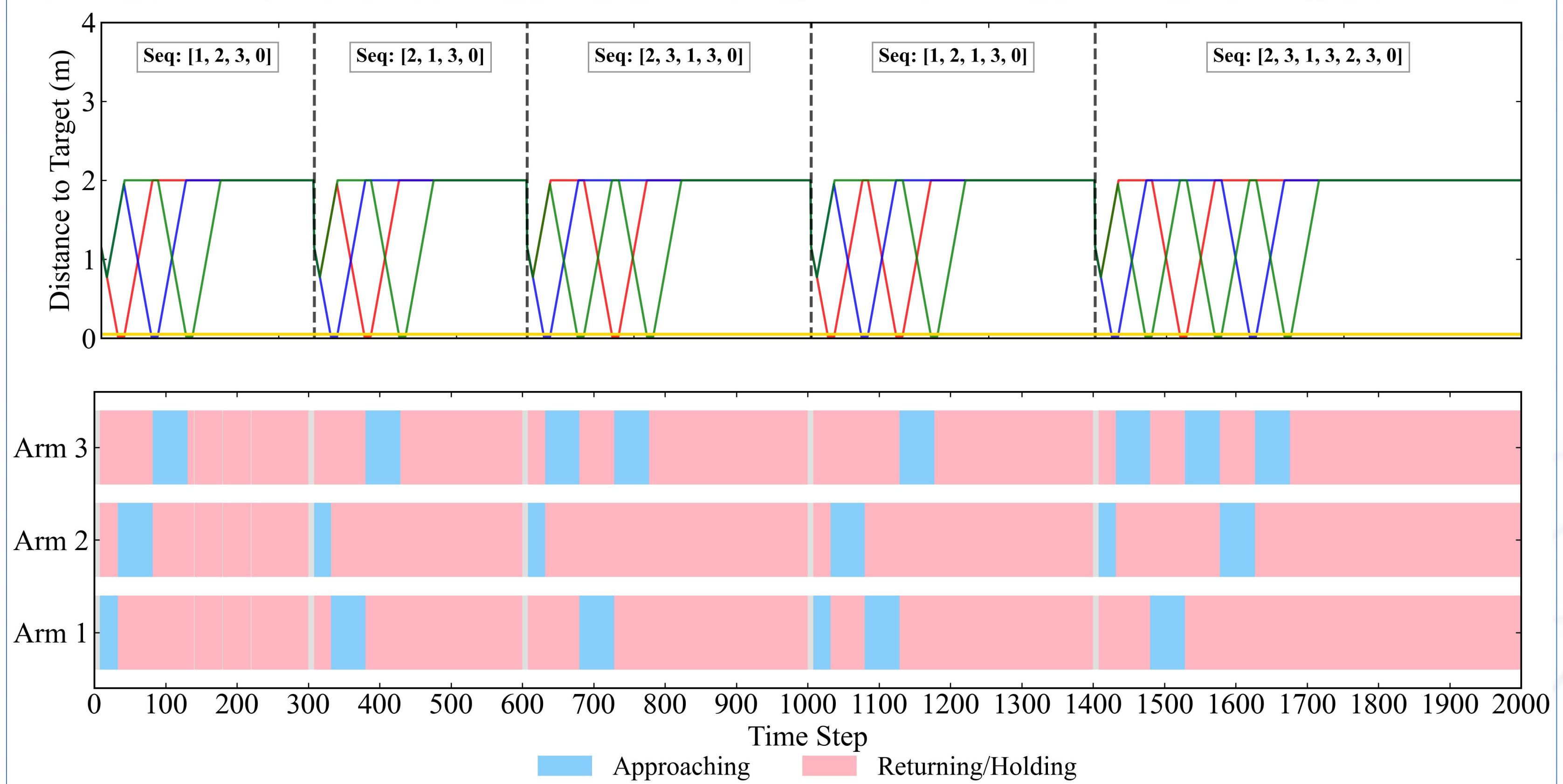


Continuous Long-Horizon Adaptation:

The system seamlessly adapts to abrupt shifts in instruction logic without catastrophic forgetting or collisions. Trajectory corrections occur within just a few gradient steps.

Robust Negotiated Turn-Taking:

Instead of relying on abstract task states, the multi-arm system coordinates precise interleaved activations by directly tracking specific robotic arm identifiers. This ensures exact execution of complex sequential orders.



⑤ Efficiency

Difficulty	Input Type	Time (s)	Accuracy (%)
Easy	Audio	0.008±0.001	93.80±1.70
	Audio (noise)	0.008±0.001	87.40±2.70
	Audio-Visual	0.389±0.038	94.60±0.80
	Audio-Visual (noise)	0.498±0.058	92.80±2.14
Medium	Audio	0.015±0.001	87.80±1.60
	Audio (noise)	0.017±0.001	77.20±2.00
	Audio-Visual	0.465±0.047	91.20±2.04
	Audio-Visual (noise)	0.571±0.034	88.00±3.74
Hard	Audio	0.023±0.002	83.00±1.50
	Audio (noise)	0.025±0.003	67.00±3.50
	Audio-Visual	0.489±0.028	90.60±2.24
	Audio-Visual (noise)	0.516±0.079	83.20±1.60

Instruction generation remains strictly sub-second (under 0.6s) across all task complexities and noise levels, fully satisfying real-time control requirements.

⑥ Future Works

Sim-to-Real Transfer:

Validate the system on physical UR5 robotic arms, testing robustness under real-world communication delays.

Scale Extension:

Extend the framework to **larger-scale multi-robot systems (>3 arms)** to verify scalability.